SORTING AND SEARCHING ALGORITHMS

Week 6

Most common tasks in computer programs

Are **sorting** data into a particular order and **searching** for items of information.

SORTING ALGORITHMS

BUBBLE SORT: When data is meant to be sorted, different items must be compared with each other and moved so that they are in either ascending order or descending order.

4	2	6	1	3	Items 1 and 2 must be swapped.
2	4	6	1	3	Items 1 and 2 are swapped.
2	4	6	1	3	Items 2 and 3 are already in ascending order.
2	4	6	1	3	Items 3 and 4 must be swapped.
2	4	1	6	3	Items 3 and 4 have been swapped.
2	4	1	6	3	Items 4 and 5 must now be swapped.
2	4	1	3	6	Items 4 and 5 have been swapped.

2	4	1	3	6	Items 1 and 2 are in correct order.
2	4	1	3	6	Items 2 and 3 must be swapped.
2	1	4	3	6	Items 2 and 3 have been swapped.
2	1	4	3	6	Items 3 and 4 must be swapped.
2	1	3	4	6	Items 3 and 4 have been swapped.
2	1	3	4	6	Items 4 and 5 do not need to be swapped.

2	1	3	4	6	Items 1 and 2 must be swapped.
1	2	3	4	6	Items 1 and 2 have been swapped.
1	2	3	4	6	All items are now in the correct order.

Bubble sort algorithm: written description.

- 1. Start at the beginning of the list.
- 2. Compare the values in position 1 and position 2 in the list - if they are not in ascending order then swap them.
- 3. Compare the values in position 2 and position 3 in the list and swap if necessary.
- 4. Continue to the end of the list.
- 5. If there have been any swaps, repeat steps 1 to 4.

▶ **Figure 1.9** Bubble sort algorithm written as a flowchart

